CSET485 AI and Society – Assignment 1

Roll number: S24CSEU1192 | **Dataset:** ProPublica COMPAS `compas-scores-two-years.csv` | **random_state:** 192, 70/30 train/test split.
All numbers are produced by `analysis.ipynb` (executed with nbconvert) and read from `results.json`; nothing is typed by hand.

1. Preprocessing

The raw file has 7214 rows × 53 columns. **Missing values:** the columns with missing values are listed below. Most are violent-recidivism or re-offence fields that are empty by design when no event happened. `days_b_screening_arrest` is missing for 307 rows, which the filter drops. After filtering, the modelling columns have 0 missing values, so no imputation was needed.

	missing (raw)
violent_recid	7214
vr_offense_date	6395
vr_charge_degree	6395
vr_case_number	6395
vr_charge_desc	6395
c_arrest_date	6077
r_jail_out	4898
r_jail_in	4898
r_days_from_arrest	4898
r_charge_desc	3801
r_offense_date	3743
r_charge_degree	3743
r_case_number	3743
c_offense_date	1159
days_b_screening_arrest	307
c_jail_out	307
c_jail_in	307
in_custody	236
out_custody	236
c_charge_desc	29
c_case_number	22
c_days_from_compas	22

ProPublica filters (justification): ±30 days between arrest and screening, so that the COMPAS score belongs to the charge we observe; `is_recid` ≠ −1, because −1 means no case record and an unknown outcome; `c_charge_degree` ≠ `'O'`, because ordinary traffic offences carry no jail time; `score_text` ≠ `'N/A'`, because these rows have no valid score. This two-year file was already partly cleaned by ProPublica, so in practice only the first filter removes rows: 7214 → 6172.

	rows remaining
raw	7214
days_b_screening_arrest in ±30	6172
is_recid != -1	6172
c_charge_degree != 'O'	6172
score_text != 'N/A'	6172

Encoding: one-hot with reference categories: sex (Female), race (Caucasian; Asian and Native American merged into Other), charge degree (Misdemeanour). Race counts: African-American 3175,

Caucasian 2103, Hispanic 509, Other 385. **Scaling:** the numeric features (age, priors_count, juv_fel/misd/other counts) are standardised with a StandardScaler fitted on the training set only. **Outliers** (above Q3 + 1.5·IQR): age 51, priors_count 492, juv_fel_count 208, juv_misd_count 352, juv_other_count 461. They are kept, because these are real criminal histories, the skew is in the right tail, and removing them would exclude exactly the people the tool is aimed at. **Features** (identical for both models, no score columns): age, priors_count, juv_fel_count, juv_misd_count, juv_other_count, sex_Male, race dummies, charge_Felony. Split: 4320 training rows / 1852 test rows. Two-year recidivism base rate: 45.5%.

2. Part 1 – Model A (OLS) vs Model B (logistic)

Model A: OLS predicting `decile_score` ($R^2 = 0.442$). Model B: logistic regression predicting `two_year_recid`, the **"is_recidivist"** outcome (1 = re-arrested within two years). On the test set, A is thresholded at its median prediction **4.480** (≥ median = high risk) and B at p ≥ 0.5. The median of B's probabilities is 0.447.

	B low (p < 0.5)	B high (p ≥ 0.5)
A low (< median)	901	25
A high (≥ median)	226	700

> **Mismatch:** 251 of 1852 = **13.55%**. A flags 50.0% as high risk and B flags 39.1%. Correlation between A's predictions and B's probabilities: 0.954. Accuracy against two_year_recid: A 64.5%, B 69.1%.

Why they differ: the ranking is almost the same, but the cuts are not. A's median is a relative cut, so it labels 50% as high risk by construction. B's 0.5 is an absolute probability, and because the base rate is below 50% it labels fewer people. In addition, A learns COMPAS's scoring, which puts much more weight on race (0.541 decile points) than the real outcome does in B (0.003 log-odds).

	Model A OLS (decile pts)	Model B Logit (log-odds)
const	3.921	-0.558
age	-1.167	-0.502
priors_count	1.240	0.707
juv_fel_count	0.117	-0.002
juv_misd_count	0.082	0.006
juv_other_count	0.165	0.116
sex_Male	-0.000	0.362
race_AfricanAmerican	0.541	0.003
race_Hispanic	-0.459	-0.200
race_Other	-0.808	-0.238
charge_Felony	0.465	0.213

3. Part 2 – Statistical analyses

2.1 Simpson's paradox

Logistic models for two_year_recid were fitted on the full filtered data (n = 6172) and within subgroups. Cells show coefficient [95% CI] and p-value; numeric features are per 1 SD. Split 1 compares African-American (subgroup A) with Caucasian (subgroup B) defendants:

predictor	Full	African-American	Caucasian	diff p (AA vs C)	sign reversal
age	-0.506 [-0.570, -0.442] p=<0.001	-0.594 [-0.694, -0.495] p=<0.001	-0.435 [-0.532, -0.337] p=<0.001	0.025	False
priors_count	0.760 [0.683, 0.837] p=<0.001	0.717 [0.621, 0.813] p=<0.001	0.916 [0.752, 1.079] p=<0.001	0.040	False
juv_fel_count	0.032 [-0.051, 0.115] p=0.449	0.030 [-0.066, 0.126] p=0.540	-0.023 [-0.200, 0.153] p=0.794	0.602	True
juv_misd_count	-0.006 [-0.081, 0.069] p=0.873	-0.049 [-0.135, 0.037] p=0.266	0.036 [-0.140, 0.212] p=0.689	0.395	True
juv_other_count	0.125 [0.061, 0.189] p=<0.001	0.256 [0.152, 0.360] p=<0.001	0.012 [-0.079, 0.104] p=0.793	<0.001	False
sex_Male	0.333 [0.192, 0.475] p=<0.001	0.522 [0.320, 0.724] p=<0.001	0.086 [-0.140, 0.311] p=0.456	0.005	False
charge_Felony	0.222 [0.107, 0.338] p=<0.001	0.187 [0.022, 0.351] p=0.026	0.188 [-0.006, 0.383] p=0.058	0.991	False

Split 2 compares male (subgroup A) with female (subgroup B) defendants:

predictor	Full	Male	Female	diff p	sign reversal
age	-0.506 [-0.570, -0.442] p=<0.001	-0.526 [-0.596, -0.455] p=<0.001	-0.419 [-0.571, -0.266] p=<0.001	0.214	False
priors_count	0.760 [0.683, 0.837] p=<0.001	0.717 [0.636, 0.799] p=<0.001	1.082 [0.838, 1.325] p=<0.001	0.005	False
juv_fel_count	0.032 [-0.051, 0.115] p=0.449	0.040 [-0.045, 0.125] p=0.362	-0.066 [-0.538, 0.405] p=0.784	0.666	True
juv_misd_count	-0.006 [-0.081, 0.069] p=0.873	-0.010 [-0.087, 0.067] p=0.804	0.016 [-0.251, 0.283] p=0.906	0.855	True
juv_other_count	0.125 [0.061, 0.189] p=<0.001	0.128 [0.059, 0.197] p=<0.001	0.072 [-0.108, 0.252] p=0.432	0.565	False
race_AfricanAmerican	0.063 [-0.061, 0.187] p=0.318	0.156 [0.017, 0.294] p=0.027	-0.313 [-0.598, -0.027] p=0.032	0.004	True
race_Hispanic	-0.174 [-0.389, 0.040] p=0.111	-0.160 [-0.393, 0.074] p=0.181	-0.130 [-0.669, 0.408] p=0.635	0.923	False
race_Other	-0.192 [-0.432, 0.048] p=0.117	-0.100 [-0.360, 0.160] p=0.451	-0.632 [-1.288, 0.025] p=0.059	0.140	False
charge_Felony	0.222 [0.107, 0.338] p=<0.001	0.168 [0.040, 0.296] p=0.010	0.414 [0.139, 0.689] p=0.003	0.112	False

Finding: the race coefficient **reverses sign** across sex. It is 0.063 pooled (p = 0.318), +0.156 among men (p = 0.027) and -0.313 among women (p = 0.032); the difference test gives p = 0.004. In the race split no significant predictor reverses sign, but the effects differ significantly: sex_Male 0.522 vs 0.086 (p = 0.005), juv_other_count (p = <0.001), priors (p = 0.040) and age (p = 0.025).
Policy implication: a pooled model averages away opposite effects in subgroups, so validation by subgroup and by intersection (e.g. sex × race) should be required before any claim that the tool is "race-neutral".

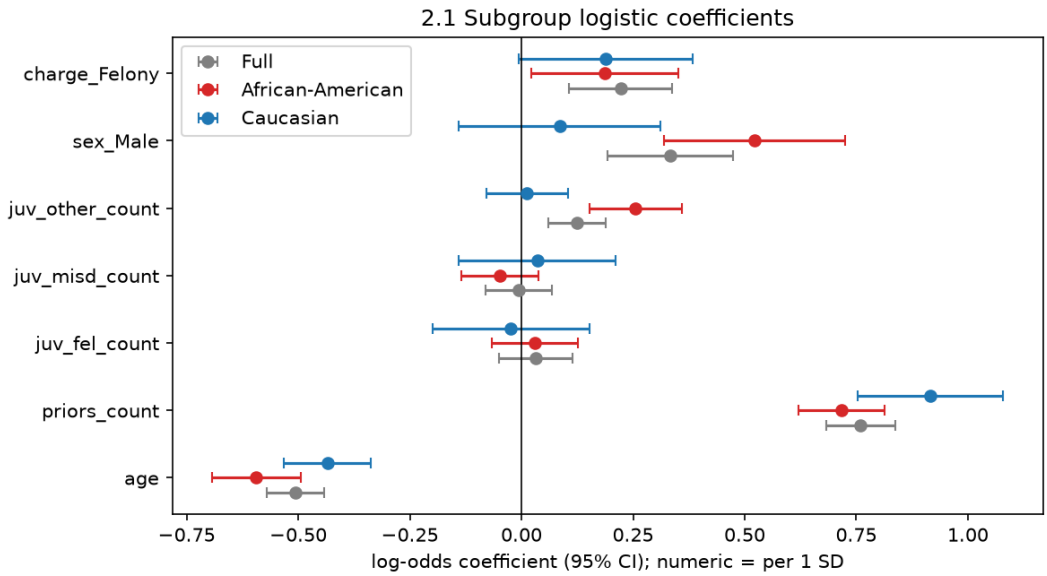


Figure 1. Coefficients with 95% CIs: full data vs African-American vs Caucasian.

2.2 Omitted-variable bias (dropping priors_count)

	full	reduced	abs_change	% change	full_p	reduced_p
predictor						
age	-0.502	-0.322	0.180	35.923	0.000	0.000
juv_fel_count	-0.002	0.149	0.151	8312.484	0.969	0.004
juv_misd_count	0.006	0.165	0.158	2533.733	0.890	0.000
juv_other_count	0.116	0.166	0.051	43.832	0.003	0.000
sex_Male	0.362	0.495	0.133	36.839	0.000	0.000
race_AfricanAmerican	0.003	0.251	0.248	8369.453	0.969	0.000
race_Hispanic	-0.200	-0.179	0.021	10.478	0.124	0.154
race_Other	-0.238	-0.294	-0.056	-23.497	0.103	0.039
charge_Felony	0.213	0.354	0.141	66.022	0.002	0.000

The **biggest shift** is in **race_AfricanAmerican**: 0.003 → 0.251 (+8369.5%), going from not significant to p = <0.001. The juvenile counts also become significant. Age weakens by 35.9%, and charge_Felony rises by 66.0%. AIC goes from 5329.0 to 5636.5. Percentage changes are inflated where the full coefficient is close to 0, so the absolute changes are shown too.

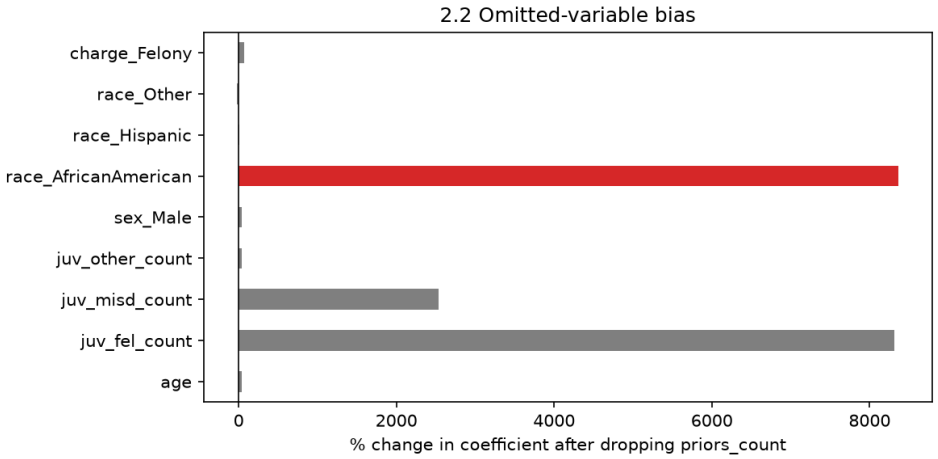


Figure 2. % change in each coefficient after omitting priors_count.

2.3 Adversarial (confidently wrong) subset

Rows with p ≥ 0.75 and y = 0, or p ≤ 0.25 and y = 1: **57** (3.08% of the test set). 23 are false-high and 34 are false-low. A sample of 50 was drawn with seed 192 (21.0 false-high, 29.0 false-low).

	Full test	Adversarial (all)	Sample of 50
n	1852.00	57.00	50.00
% African-American	50.97	50.88	54.00
% Caucasian	34.29	35.09	32.00
% Male	80.13	78.95	78.00
mean age	34.15	46.63	46.36
% age<25	22.52	7.02	4.00
% age>45	18.52	57.89	56.00
mean priors	3.12	5.72	6.18
% priors=0	34.61	31.58	34.00
% priors>=5	23.33	35.09	38.00
% felony	65.50	57.89	56.00
% y=1 (recidivated)	44.17	59.65	58.00
mean p	0.46	0.44	0.45

Shared trait: the subset is older and sits at the extremes of the age and priors axes. 56.0% are over 45, against 18.5% in the test set, and 96% of the 50 have an extreme profile: confidently-high cases with 5+ priors or under 25, or confidently-low cases aged 35+ with at most one prior. 90.5% of the false-highs have 5+ priors, and 100% of the false-lows are aged 35+. **Why the model fails:** the log-odds are linear in priors (0.707/SD) and age (-0.502/SD), so extreme values push p to the tails, and the 8 features carry no information about desistance or context that could pull it back.

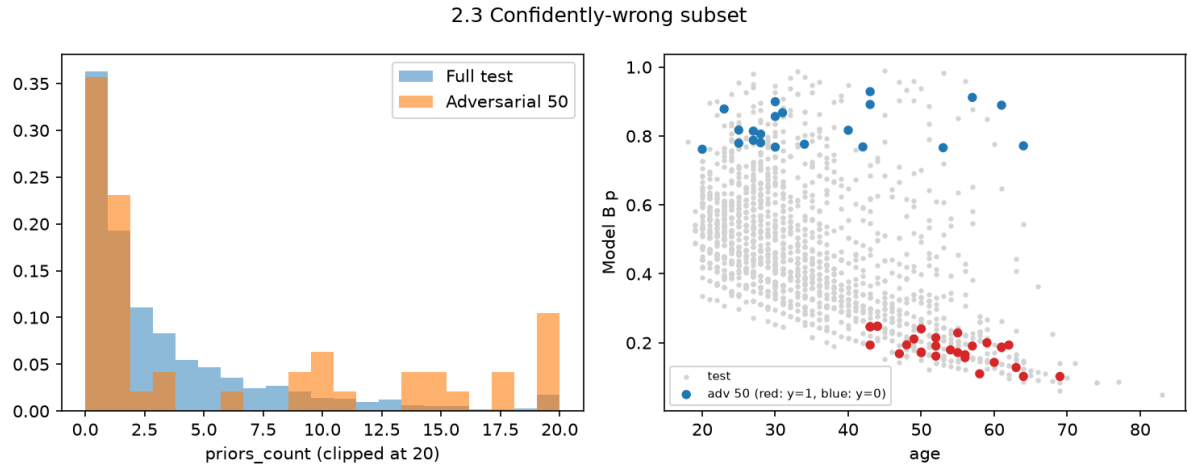


Figure 3. Priors distribution and age vs predicted probability for the adversarial sample.

2.4 Fairness threshold sweep (A = African-American n=944, B = Caucasian n=635, test set)

	t	DP	EO	PP	acc	acc_A	acc_B	sel_A	sel_B	TPR_A	TPR_B	FPR_A	FPR_B	prec_A	prec_B
0	0.050	1.000	1.000	1.501	0.442	0.530	0.353	1.000	1.000	1.000	1.000	1.000	1.000	0.530	0.353
1	0.150	1.052	1.010	1.441	0.467	0.532	0.398	0.994	0.945	0.996	0.987	0.991	0.922	0.531	0.368
2	0.250	1.195	1.053	1.324	0.536	0.556	0.507	0.950	0.795	0.978	0.929	0.919	0.723	0.545	0.412
3	0.350	1.448	1.186	1.230	0.613	0.603	0.611	0.855	0.591	0.932	0.786	0.768	0.484	0.577	0.469
4	0.450	1.769	1.495	1.269	0.668	0.665	0.657	0.646	0.365	0.794	0.531	0.480	0.275	0.651	0.513
5	0.550	2.297	1.893	1.263	0.684	0.681	0.683	0.401	0.175	0.578	0.299	0.203	0.107	0.763	0.604
6	0.650	2.963	2.010	1.019	0.645	0.603	0.690	0.219	0.074	0.332	0.165	0.092	0.024	0.802	0.787
7	0.750	4.036	2.867	1.067	0.610	0.553	0.665	0.121	0.030	0.192	0.067	0.041	0.010	0.842	0.789
8	0.850	5.830	4.032	1.038	0.583	0.511	0.654	0.055	0.009	0.090	0.022	0.016	0.002	0.865	0.833
9	0.950	2.691	1.792	1.000	0.564	0.479	0.650	0.008	0.003	0.016	0.009	0.000	0.000	1.000	1.000

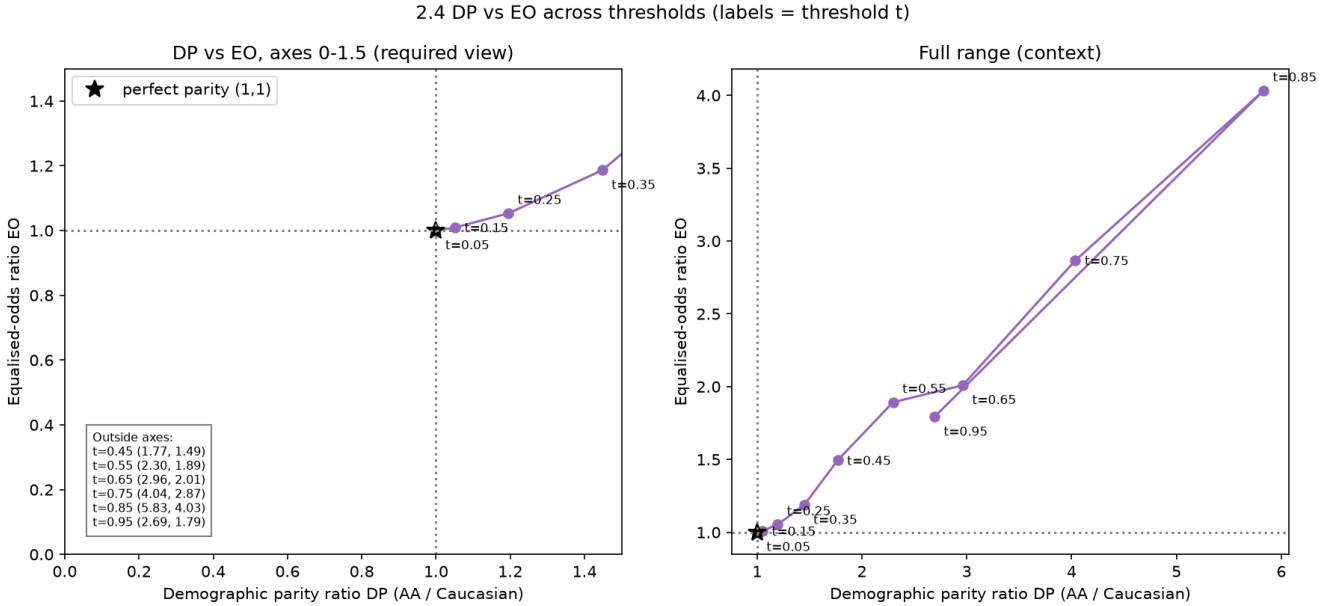


Figure 4. DP (x) vs EO (y). Left: required 0–1.5 axes, points labelled by threshold, with off-axis thresholds listed in the box. Right: full range.

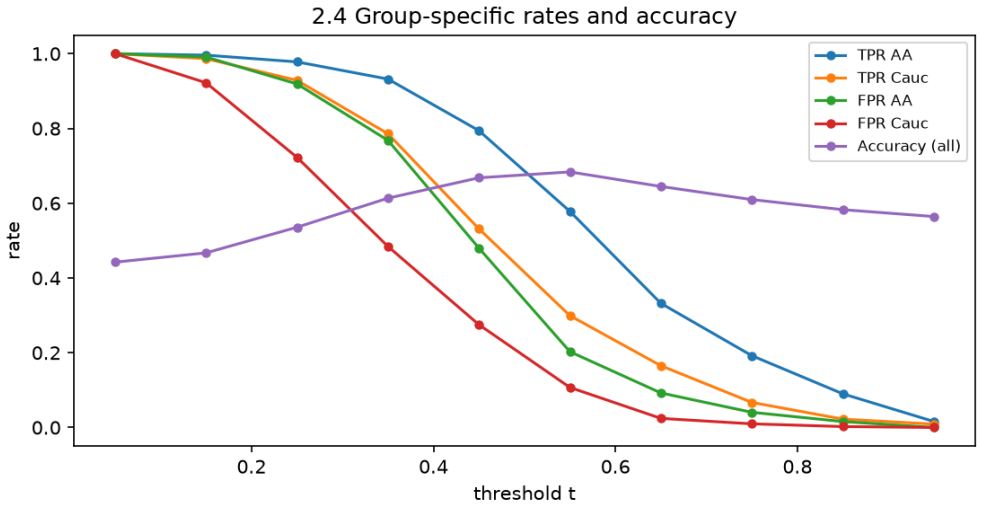


Figure 5. TPR, FPR by group and overall accuracy across thresholds.

Trade-off (baseline t = 0.5: DP 2.109, EO 1.739, PP 1.238, accuracy 69.1%). Fairness is measured as closeness to parity, min(r, 1/r). Z = % gain in DP closeness and W = % loss in EO closeness (a negative W means EO also improves).

	DP_from	DP_to	Z	EO_from	EO_to	W	cost	PP_to	W_pp	acc_to	W_acc	cost_acc
t												
0.15	2.109	1.052	100.531	1.739	1.010	-72.261	-0.719	1.441	14.098	0.467	32.369	0.322
0.25	2.109	1.195	76.495	1.739	1.053	-65.112	-0.851	1.324	6.449	0.536	22.439	0.293
0.35	2.109	1.448	45.677	1.739	1.186	-46.606	-1.020	1.230	-0.638	0.613	11.181	0.245
0.45	2.109	1.769	19.232	1.739	1.495	-16.354	-0.850	1.269	2.414	0.668	3.284	0.171

> Improve DP from 2.11 to 1.19 (76.5% gain) and EO goes from 1.74 to 1.05 (W = -65.1%, so it is not a loss), giving cost = W/Z = -0.85. For t = 0.15: DP 2.11 → 1.05 (100.5% gain), EO 1.74 → 1.01 (W = -72.3%), cost = -0.72. With a single global threshold, DP and EO move together. The real costs are **accuracy** (22.4% loss at 0.25, cost 0.29) and **predictive parity** (PP 1.24 → 1.44, 14.1% loss at 0.15).

4. Part 3 – Discussion questions

Q1. Why do Models A and B disagree? (mismatch; median of linear predictions vs 0.5)

Model A (OLS on decile_score) and Model B (logistic on two_year_recid, the "is_recidivist" label) were fitted on the same 4320 training rows with identical features, yet on the 1852 test rows they disagree on 251 people, a mismatch of 13.55%. The agreement matrix shows 901 low/low, 700 high/high, 226 cases that A calls high but B calls low, and only 25 cases the other way. The asymmetry is the whole story. A's threshold is its median linear prediction, 4.480 decile points, so by construction exactly 50.0% of the test set is "high risk". B's threshold is a probability of 0.5, but the median of B's predicted probabilities is only 0.447, so $p \geq 0.5$ flags just 39.1%. In effect B cuts at the 60.9th percentile while A cuts at the 50th. The two rankings are almost the same (correlation between A's predictions and B's probabilities = 0.954), so the disagreement comes mainly from where each cut sits, not from different views of who is riskier. The two thresholds also live on different scales. A's median is relative: it adapts to whatever the sample looks like and has no link to the base rate. B's 0.5 is absolute and tied to the observed recidivism base rate of 45.5%, which is below 50%. There is a second, smaller source: the targets differ. The decile score is COMPAS's own judgement (OLS $R^2 = 0.442$), and it puts much more weight on race than the outcome does. The African-American coefficient is 0.541 decile points in A against 0.003 log-odds in B. Against real outcomes, A's labels are 64.5% accurate and B's are 69.1%. So the 226 "A-high, B-low" people are mostly people whom the score-imitating model ranks above the median but whose actual re-arrest probability is below one half.

[295 words]

Q2. Subgroup coefficients, CIs and the policy change

Two splits were tested. By sex, the race effect reverses sign, which is a Simpson-type pattern. In the full-data logistic model the African-American coefficient is 0.063 (95% CI -0.061 to 0.187, p = 0.318), so it is not significant. Among men (n = 4997) it is positive at 0.156 (CI 0.017 to 0.294, p = 0.027). Among women (n = 1175) it is negative at -0.313 (CI -0.598 to -0.027, p = 0.032). The two subgroup coefficients differ significantly (z = 2.89, p = 0.004). Pooling hides opposite associations because men make up 81.0% of the data and drive the aggregate. By race (African-American n = 3175 vs Caucasian n = 2103), no significant predictor flips sign; only the null juvenile-felony and misdemeanour terms do. The effects do differ in size, though. Being male carries log-odds of 0.522 (CI 0.320 to 0.724, p <0.001) for African-American defendants but only 0.086 (CI -0.140 to 0.311, p = 0.456) for Caucasians (difference p = 0.005). Priors per SD are 0.717 vs 0.916 (difference p = 0.040), and "other" juvenile offences are 0.256 vs 0.012 (difference p = <0.001). Policy implication: one pooled coefficient vector is a misleading description of every group. A tool that says "race adds nothing" (pooled p = 0.318) is quietly averaging an upward shift for men with a downward one for women. Auditors should require models to be validated by subgroup, including intersections such as sex × race, before deployment. Where effects truly differ, the agency should publish group-specific calibration rather than one score, and it should not read a non-significant pooled race term as evidence that the tool is race-neutral.

[273 words]

Q3. The dropped variable: what moved, and the biggest shift

I dropped priors_count from Model B. It is the strongest predictor in the full model (log-odds 0.707 per SD), and it is correlated with race (r = 0.215 with the African-American dummy), sex (r = 0.119), felony charge (r = 0.145) and juvenile records (r = 0.268 for misdemeanours). That makes it a textbook confounder. Coefficients change from full to reduced as follows (% change relative to |full|; the full table is in Part 2.2): age: -0.502 → -0.322 (+35.9%); sex_Male: 0.362 → 0.495 (+36.8%); race_AfricanAmerican: 0.003 → 0.251 (+8369.5%); charge_Felony: 0.213 → 0.354 (+66.0%); juv_other_count: 0.116 → 0.166 (+43.8%). The biggest shift is race_AfricanAmerican: it rises from a null 0.003 (p = 0.969) to 0.251 (p = <0.001), a +8369% change. The percentage looks extreme because

the base is close to zero. The substantive point is that a race effect appears from nothing once criminal history is omitted. The juvenile felony and misdemeanour coefficients behave the same way (+8312% and +2534%), becoming significant proxies for priors. Age weakens by 35.9% because younger people have fewer priors, so age was partly suppressing the priors signal. Model fit falls (AIC 5329.0 → 5636.5). The direction matches the omitted-variable-bias formula: bias = β_priors × the regression of priors on each variable. β_priors is positive and priors correlate positively with being African-American, so the race coefficient is pushed upward. The lesson for policy is that whether a model "uses race" depends on what else it includes. Removing a legitimate covariate can make a protected attribute look predictive, and keeping it can hide the fact that the covariate itself carries the effects of policing that differ by race.

[274 words]

Q4. A concrete real-world method for detecting a hidden confound

A concrete method is a negative-control and proxy audit, run at the start of any deployment. Step 1: list plausible unmeasured confounders for this setting, such as policing intensity in the defendant's neighbourhood, poverty, and access to a lawyer. Step 2: link available proxies to the records, for example census-tract arrest rates per capita or public-defender status taken from court dockets. Step 3: refit the model with and without each proxy and track the coefficients of the protected attributes, as in Part 2.2. There, adding or removing a single covariate moved the African-American coefficient from 0.003 to 0.251, so movements of that size are detectable. Step 4: add a negative-control outcome, meaning something the features should not cause but the confounder would, such as being arrested for an offence later dismissed for lack of evidence. If "race" or "priors" predict dismissed arrests as strongly as convictions, the signal is coming from enforcement, not behaviour. Step 5: run a sensitivity analysis (Rosenbaum bounds or E-values) to ask how strong an unseen confounder would have to be to explain away the race association. Step 6: check heterogeneity. Part 2.1 showed that the pooled race coefficient 0.063 hides 0.156 for men and -0.313 for women. Unexpected sign changes across strata are a warning sign that something omitted varies by group. In Broward County this could be done with Florida court data: compute re-arrest rates per officer-hour by district, add them as a covariate, and see whether the African-American priors gradient (0.717 vs 0.916 per SD in the subgroup models) shrinks. If it does, part of what "priors" measures is exposure to policing, a hidden confound that a recidivism tool would otherwise turn into an individual risk label.

[284 words]

Q5. The 50-row adversarial subset: shared trait and failure mechanism

The adversarial subset holds every test row where Model B is confidently wrong: 57 people, or 3.08% of the 1852 test rows. Of these, 23 had $p \geq 0.75$ but did not reoffend, and 34 had $p \leq 0.25$ but did. A sample of 50 drawn with seed 192 (21.0 false-high and 29.0 false-low) differs sharply from the test set. The race mix is similar (54.0% vs 51.0% African-American) and so is the share of men (78.0% vs 80.1%). Age is not. Mean age is 46.4 vs 34.2, and 56.0% are older than 45 compared with 18.5% in the test set. Only 4.0% are under 25 (test set 22.5%). The shared trait is an extreme position on the model's two dominant axes. 96% of the 50 are either confidently-high people with 5+ priors or under 25, or confidently-low people aged 35+ with at most one prior. Within the groups, 90.5% of the false-highs have 5+ priors, and 100% of the false-lows are aged 35+ (82.8% over 45). The mechanism follows from the coefficients. Log-odds are linear in standardised priors (0.707/SD) and age (-0.502/SD). Priors have SD 4.78 (mean 3.30), so someone with 15 priors sits 2.45 SD out and gains 1.73 log-odds from priors alone, which pushes p above 0.75 whatever else is true. Older age pushes p below 0.25 in the same way. The model has no features for desistance (people who stop offending), employment or supervision, so it cannot tell which heavy-history or older people differ from the average. Its extreme probabilities are extrapolations along one direction, and the 8 features carry no information to correct them.

[269 words]

Q6. Two people from different groups under different thresholds

Consider two 23-year-old men facing the same felony charge, each with one prior. One is African-American, the other Caucasian. Under Model B their probabilities are almost identical, because the race coefficient is small (0.003 log-odds). Now suppose a court picks a threshold based on group statistics. At the default t = 0.5, the test set flags 54.1% of African-American defendants and 25.7% of Caucasians (DP = 2.11). Among people who did not reoffend, 33.8% of African-Americans are wrongly flagged versus 17.0% of Caucasians. A reform committee might then propose group-specific cut-offs to equalise false positives, with roughly t = 0.65 for African-Americans (FPR 9.2%) and t = 0.45 for Caucasians (FPR 27.5%). Under that rule, if both men score p = 0.55, the African-American man is released and the Caucasian man is detained. Same record, same probability, opposite decision, purely because of group membership. That is disparate treatment, and it can be illegal. Under a single threshold, the African-American man faces the base-rate-driven disparity instead. His group's observed re-arrest rate in the test set is 53.0% against 35.3% for Caucasians, so more members of his group sit near the cut, and his group's FPR at t = 0.5 is 1.98× higher. Neither rule treats both men "fairly" in every sense. Group thresholds equalise error rates but break individual consistency; a single threshold keeps consistency but produces unequal error burdens. The concrete harm also differs. A false positive means pretrial detention, with possible job loss and pressure to plead guilty. A false negative is a missed re-arrest, whose cost falls on the public. Any choice should be made explicitly and recorded in writing, not left buried in a default.

[278 words]

Q7. Defending one threshold, and the strongest counter-argument

I recommend a single threshold of t = 0.5, the same for everyone. Evidence: (1) it gives the highest test accuracy of all thresholds evaluated, 69.1%, against 66.8% at 0.45, 68.4% at 0.55 and 53.6% at 0.25. (2) Precision is reasonably high in both groups (70.6% for African-American and 57.1% for Caucasian defendants), so most people flagged as high risk do in fact reoffend. The PP ratio is 1.24, compared with 1.44 at t = 0.15. (3) A logistic model's 0.5 cut-off has a plain meaning, "more likely than not", which can be explained in court and challenged on appeal. A tuned threshold such as 0.25 also flags 79.5% of Caucasians and 95.0% of African-Americans, which amounts to detaining almost everyone. (4) Lowering the threshold to move DP towards parity costs accuracy: going to 0.25 reduces accuracy by 22.4%. The strongest counter-argument is about error-rate fairness. At t = 0.5 the false-positive rate for African-American defendants who never reoffend is 33.8%, against 17.0% for Caucasians, a ratio of 1.98. DP is 2.11: African-Americans are flagged at 2.1× the Caucasian rate. Because the cost of a false positive is liberty, a critic can reasonably argue that accuracy which falls unequally on one group is not a neutral virtue. The 7.7-point accuracy edge over t = 0.35 (DP 1.45) is being paid for disproportionately by Black defendants. That critic would also note that the base-rate gap (53.0% vs 35.3%) partly reflects differences in policing, so "calibrated" may itself encode bias. My defence holds only if t = 0.5 is paired with human review of every flagged case and regular monitoring of the FPR ratio.

[272 words]

Q8. DP vs EO for this context: trade-off and what is sacrificed

In pretrial or parole decisions, I prefer equalised odds (EO) to demographic parity (DP). DP asks for equal flag rates whatever the true outcomes; EO asks that people who will not reoffend, and people who will, face equal error rates regardless of race. Because the harm of a false positive is detention, the FPR component of EO maps directly onto the injustice that matters. At t = 0.5 the test set gives DP = 2.11 and EO = 1.74 (TPR ratio 1.74, FPR ratio 1.98), so both are far from parity. Measuring fairness as closeness to 1, min(r, 1/r): moving to t = 0.25 improves DP from 2.11 to 1.19 (76.5% gain), and EO goes from 1.74 to 1.05, a W = -65.1% "loss" (negative, so in fact a gain), giving cost = W/Z = -0.85. Going to t = 0.15 gives DP 1.052 (100.5% gain) and EO 1.010 (W = -72.3%, cost -0.72). So in this data a single threshold never trades DP against EO. Both approach 1 only because the threshold falls so low that almost everyone is flagged. What is actually sacrificed is accuracy and predictive parity. Accuracy falls from 69.1% to 53.6% at 0.25 (a 22.4% loss, cost 0.29 per unit of DP gain) and to 46.7% at 0.15.

Predictive parity worsens from 1.24 to 1.44 at 0.15 (14.1% loss). The Caucasian precision falls to 0.37, so a "high-risk" label means something different for each group. Fairness achieved by flagging everyone is meaningless: liberty is sacrificed equally rather than allocated fairly. With base rates of 53.0% vs 35.3%, the impossibility result (Kleinberg, Chouldechova) applies: calibration and equal error rates cannot both hold. Real EO therefore needs group-aware post-processing or better features, not a moved global threshold.

[290 words]

5. Part 4 – Reflection: can one number certify a fair risk tool?

Direct answer

No. A single accuracy figure, such as this model's 69.1% at t = 0.5, cannot tell us whether a recidivism tool is fit to use on people, because it hides who bears the errors, how the threshold was chosen, and what the label actually measures.

Why a single number fails

First, aggregate accuracy averages over groups whose error profiles are very different. At the same t = 0.5, 33.8% of African-American defendants who never reoffended were flagged high risk, compared with 17.0% of Caucasians. Accuracy within the two groups barely differs (69.4% vs 68.3% at t = 0.5), so a headline number cannot reveal a near-doubling of wrongful flags. The Simpson analysis makes the same point about coefficients. The pooled race effect (0.063, p = 0.318) looks negligible, yet among men it is 0.156 and among women -0.313, and both are significant. One summary averaged two opposite truths.

Second, any number depends on a threshold that someone chose. Model A and Model B rank people almost identically (correlation 0.954), yet they disagree on 13.55% of cases only because A cuts at its median and B at 0.5. Across the sweep, accuracy ranges from 44.2% to 68.4% and DP from 1.00 to 5.83 for the same fitted model. Reporting one point on that curve hides a policy decision inside a technical default.

Third, the metric depends on how the model is specified and on what the label means. Dropping one confounder, priors_count, moved the African-American coefficient from 0.003 to 0.251. Whether the model "uses race" therefore depends on which other variables are included. The label two_year_recid records re-arrest, not reoffending, so policing intensity enters both the label and the strongest feature, prior arrests. A model can be very accurate at predicting arrests and still be unfair.

Fourth, averages hide confident failures. 57 test cases (3.08%) were wrong at probabilities beyond 0.75 or 0.25, and they cluster among older people and people with long records (56.0% of the sampled 50 were over 45, against 18.5% in the test set). For them, the model's confidence is the problem, and accuracy does not measure confidence. A judge who sees "p = 0.80" reads it as near-certainty, but for a long-record defendant in this region of the data it is an extrapolation along a single coefficient, not evidence about that individual.

Grounding in numbers

Concretely, the 69.1% accuracy sits alongside DP = 2.11, EO = 1.74 and PP = 1.24. Pushing DP to 1.05 would require t = 0.15, where accuracy falls to 46.7%. Base rates differ (53.0% vs 35.3%), so no single threshold can satisfy all criteria at once. Meanwhile 57 people were confidently misjudged, and removing priors_count changed the race coefficient by +8369%.

Closing

A responsible audit reports a dashboard, not a score. It should cover error rates by group and by intersection, the full threshold curve, how coefficients hold up when variables are added or removed, and the confident failures, together with a plain statement of which fairness criterion was chosen and who pays for that choice in liberty.

[Part 4 total: 498 words; sections: Direct answer 45, Why a single number fails 336, Grounding in numbers 61, Closing 56]